

Original articles

Isoparametric formulation for a quadrilateral element described by non-differentiable functions using Gaussian points[☆]Esmaeil Motevali¹, Houshyar Noshad^{*,2}

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ABSTRACT

The isoparametric formulation for mapping from natural (intrinsic) coordinates to global coordinates is performed to facilitate point placement for solving differential equations using numerical methods. Here, we attempt to derive the general form of this formulation for a quadrilateral so that these shapes can be considered as an image of the reference square $\Omega_n = [-1, 1] \times [-1, 1]$ in natural coordinates with desired accuracy. Since this formulation is based on Lagrange interpolation, its error analysis is conducted based on the type of fixed points chosen. It is afterwards shown that, contrary to common belief, Legendre–Gauss points, which do not include the endpoints of the natural interval, can also be used for interpolation-based mapping. To the best of our knowledge, the application of Legendre–Gauss points in this specific context has not been previously reported. Additionally, using a practical example and its precise mapping, we demonstrate that in view of the limitations of triangular mapping, any arbitrary shape in global coordinates—even those with sides described by non-differentiable or piecewise functions—can be considered as a projection of the reference square. This, along with the definition domain of two-dimensional spectral methods as the square $[-1, 1] \times [-1, 1]$, highlights the advantage and superiority of the isoparametric formulation for the quadrilateral element.

1. Introduction

Differential equations gain physical significance when their boundary conditions are considered. On the other hand, real-world problems that these equations are designed to describe usually do not have smooth boundaries. The solution space of the equation is often surrounded by complex curves that can be described by high-degree polynomials or piecewise and non-differentiable functions. Moreover, such equations, which describe the real world, typically do not have analytical solutions, and numerical methods must be used to solve them. These methods also require discretization in the solution space, where the irregularity of the boundaries complicates this process, making the equation harder to solve. Additionally, some of these methods, such as the finite element method (FEM), use these points to mesh the solution space, which further complicates the process. Therefore, it is better to first find a way to smooth the boundaries before solving the equation.

To this end, a method has been developed that allows irregular shapes in the global coordinate system to be represented as images of regular reference domains, such as the line segment $[-1, 1]$, the square $[-1, 1] \times [-1, 1]$, or the right-angled triangle, in the

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dimensionless natural (intrinsic) coordinate system. Accordingly, the solution of the differential equation in the global coordinates can be expressed as a function of the solution in the natural coordinates. This approach, called the *isoparametric formulation*, was originally introduced to facilitate meshing in the finite element method (FEM) [1]. According to the early classical discussions of the isoparametric formulation [2], the term “isoparametric” indicates that both the geometry (in global coordinates) and the unknown field variables (e.g., displacement in solid mechanics, temperature in heat transfer, velocity in fluid flow) are interpolated using shape functions of the same polynomial degree [3]. In line with this definition, if the polynomial degree used for the geometry approximation is lower or higher, the method is referred to as the *subparametric* or *superparametric* formulation, respectively [4,5]. While these concepts originated in FEM, they are not restricted to it; rather, they provide a systematic way to regularize the solution space of differential equations, making them applicable to other numerical methods as well. In this study, we adopt the name *isoparametric formulation* to emphasize its broader applicability beyond FEM.

Furthermore, with the advancement of engineering software, even the most complex shapes can be meshed, reducing the necessity of using isoparametric formulation for the FEM. In contrast, spectral methods for solving differential equations in closed spaces use combinations of orthogonal functions such as Chebyshev and Legendre polynomials, whose domain is the natural coordinates. In other words, they always solve the differential equation on a straight line (in one dimension), a square (in two dimensions), and a cube (in three dimensions) in the natural coordinates [6]. Therefore, the use of parametric mapping for these methods is essential. Although in the FEM, regularizing the solution space of the equation makes meshing easier and reduces the number of meshes required to achieve the desired accuracy, this issue is particularly important for those who do not have access to the relevant engineering software or want to perform the meshing themselves.

Recent developments in spectral and high-order isoparametric elements have explored optimal node distributions to enhance convergence and efficiency. Bernardi, Girault, and Maday pioneered mixed spectral element approximations using Gauss–Lobatto–Legendre points for both geometry and solution interpolation, achieving exponential convergence on smooth domains [7]. Deville et al. [8] implemented high-order isoparametric spectral elements for incompressible flow with endpoint-inclusive Gauss–Lobatto nodes, and Sherwin and Karniadakis extended this framework with hp-adaptive strategies in spectral/hp elements [9].

The contribution of this study is to provide a systematic and pedagogical derivation of the isoparametric formulation for quadrilateral elements based explicitly on Lagrange interpolation. While the general concept of isoparametric mapping is well known in the finite element literature, detailed step-by-step formulations grounded in interpolation theory are rarely presented. In this work, we extract the fundamental relations in a transparent manner so that the procedure can be readily understood and reproduced. This approach is not claimed as a novelty in itself—since the pioneers of the method were certainly aware of these principles—but rather as a comprehensive and educational investigation that clarifies the underlying mathematics and makes the formulation more accessible.

In particular, we highlight the role of Legendre–Gauss points, which are traditionally excluded from isoparametric mappings due to their lack of endpoint inclusion. By carefully reformulating the interpolation, we show that these points can indeed be employed consistently within quadrilateral mappings, thereby extending their utility beyond integration and approximation in spectral methods. This perspective not only reveals the practical advantages of Legendre–Gauss points but also provides a framework that can be generalized to higher-dimensional elements. Consequently, the study serves both as a reference for practitioners and as a guide for researchers who wish to derive similar formulations for more complex geometries.

Since the isoparametric formulation is based on interpolation, this paper first reviews Lagrange interpolation and interpolation error analysis in Section 2. In Section 3, we introduce this formulation in one and two-dimensional coordinates, deriving the corresponding shape functions for beam, quadrilateral, and triangular elements. We also explain how to perform correct mapping using Legendre–Gauss points, which do not include the endpoints of the interval, in this section. Section 4 presents the necessary modifications for solving the differential equation in the natural coordinates, followed by a brief explanation of the inverse mapping from global coordinates to natural coordinates in Section 5. Finally, Section 6 concludes the paper with a practical example and related remarks, and Section 7 provides the conclusions.

2. Lagrange interpolation and interpolation error analysis

According to the Weierstrass approximation theorem, any arbitrary function such as $f(x)$ defined on the interval $[a, b]$ can be approximated by a polynomial of degree n , denoted as $P(x)$ [10]:

$$\exists \varepsilon > 0 : |f(x) - P(x)| < \varepsilon \quad \forall x \in [a, b], \quad P(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0. \quad (1)$$

Since $P(x)$ generally has $n + 1$ unknown coefficients as degrees of freedom, according to the Rouché–Capelli theorem, we need $n + 1$ independent equations to uniquely determine these coefficients [11]. These equations can be formed by equating $P(x)$ with $f(x)$ at $n + 1$ distinct points $x_0, x_1, \dots, x_n$. Thus, the unique polynomial $P(x)$ will be:

$$P(x) = \sum_{i=0}^n a_i x^i = \sum_{i=0}^n f(x_i) \mathcal{L}_{n,i}(x), \quad \mathcal{L}_{n,i}(x) := \prod_{\substack{k=0 \\ k \neq i}}^n \frac{(x - x_k)}{(x_i - x_k)}, \quad (2)$$

$$\mathcal{L}_{n,i}(x_i) = \prod_{\substack{k=0 \\ k \neq i}}^n \frac{(x_i - x_k)}{(x_i - x_k)} = 1 \quad \forall i = 0, 1, \dots, n \rightarrow P(x_i) = f(x_i) \quad \forall i = 0, 1, \dots, n. \quad (3)$$

Here, $\mathcal{L}_{n,i}(x)$ is called the Lagrange polynomial of degree n , and this polynomial approximation that interpolates any arbitrary function within its domain is known as Lagrange interpolation. Assuming $f \in C^{n+1}[a, b]$, the interpolation error in terms of the exact value of the function $f(x)$ is given by [10]:

$$f(x) = P(x) + R(x), \quad R(x) = \frac{\prod_{k=0}^n (x - x_k)}{(n+1)!} f^{(n+1)}(\xi(x)). \quad (4)$$

As is evident, if $f(x)$ is a polynomial of degree at most n , then the truncation error becomes zero. Moreover, if $f(x)$ is a polynomial of degree less than n , the degree of the interpolating polynomial $P(x)$ is reduced accordingly, and the coefficients of its higher-order terms vanish. This is the polynomial reproduction property of Lagrange interpolation: whenever the interpolation series is applied to a polynomial of degree at most n , it reproduces that polynomial exactly (see Section 3).

On the other hand, if $f(x)$ is a polynomial of degree higher than n , the error term $R(x)$ is non-zero, and the question arises under what conditions this error can be optimized. By examining Eq. (4), we observe that optimizing it is equivalent to optimizing $\omega_{n+1}(x) := \prod_{k=0}^n (x - x_k)$, since $(n+1)!$ is constant and $f^{(n+1)}(\xi(x))$ is an unknown theoretical term. Thus, only $\omega_{n+1}(x)$ can vary depending on the choice of interpolation nodes x_k . Given the continuity of $\omega_{n+1}(x)$, it is natural to employ continuous norms to measure it and to investigate under which norms the quantity $\|\omega_{n+1}(x)\|$ attains its minimum. Since $\omega_{n+1}(x)$ is a monic polynomial (i.e., a polynomial with leading coefficient 1) of degree $n+1$ ($\omega_{n+1}(x) \in \tilde{\Pi}_{n+1}$), we introduce the following lemma for further use:

Lemma 2.1. *A monic polynomial has the property that if it takes the roots of a specific polynomial of the same degree, it will be exactly equivalent to the monic form of that specific polynomial.*

If we choose the infinity norm to measure $\omega_{n+1}(x)$, we can determine the minimum value of $\|\omega_{n+1}(x)\|_\infty$ using the following theorem [10]:

Theorem 2.2. *Among all polynomials in the class $\tilde{\Pi}_n$, the monic Chebyshev polynomial $\tilde{T}_n(\eta)$ for $n \geq 1$ has the smallest l_∞ norm on $[-1, 1]$; that is*

$$\frac{1}{2^{n-1}} = \|\tilde{T}_n(\eta)\|_\infty \leq \|\omega_n(\eta)\|_\infty \quad \forall \omega_n \in \tilde{\Pi}_n, \quad -1 \leq \eta \leq 1.$$

Equality in this relation occurs only when $\omega_n(\eta) = \tilde{T}_n(\eta)$.

Based on Lemma 2.1 and Theorem 2.2, we conclude that $\omega_{n+1}(x)$ attains the smallest infinity norm on the natural interval $[-1, 1]$ when it is exactly equivalent to the monic Chebyshev polynomials. This situation occurs when the roots of $\omega_{n+1}(x)$ coincide with the roots of $\tilde{T}_{n+1}(\eta)$, which are known as the Chebyshev–Gauss (CG) points. It can be readily shown that this theorem remains valid for any closed interval, since any arbitrary closed interval such as $[a, b]$ can be mapped linearly onto $[-1, 1]$. Assuming that $x \in [a, b]$ corresponds to $\eta \in [-1, 1]$, we obtain:

$$\begin{aligned} x \longleftrightarrow \eta \Rightarrow \eta &= \frac{2(x-a)}{b-a} - 1, \quad x = \frac{(b-a)\eta + (b+a)}{2} \\ \Rightarrow \\ \omega_{n+1}(x) &= \prod_{k=0}^n (x - x_k) = \left(\frac{b-a}{2}\right)^{n+1} \prod_{k=0}^n (\eta - \eta_k). \end{aligned} \quad (5)$$

Similarly, there is a theorem with the same structure for the Euclidean norm, which allows us to determine the minimum value of $\|\omega_{n+1}(x)\|_2$:

Theorem 2.3. *Among all polynomials in the class $\tilde{\Pi}_n$, the monic Legendre polynomial $\tilde{P}_n(\xi)$ for $n \geq 1$ has the smallest l_2 norm on $[-1, 1]$; that is*

$$\left(\frac{n!}{(2n-1)!!}\right) \sqrt{\frac{2}{2n+1}} = \|\tilde{P}_n(\xi)\|_2 \leq \|\omega_n(\xi)\|_2 \quad \forall \omega_n \in \tilde{\Pi}_n, \quad -1 \leq \xi \leq 1.$$

Equality in this relation occurs only when $\omega_n(\xi) = \tilde{P}_n(\xi)$.

Similar to the discussions related to Theorem 2.2, we find that $\omega_{n+1}(x)$ has the smallest Euclidean norm when it takes the roots of $P_{n+1}(\xi)$, known as Legendre–Gauss (LG) points. These points, which have historical roots in Gaussian quadrature, do not include the endpoints of the interval and are therefore not suitable for problems requiring evaluation at the endpoints. For such cases, another set of points known as Legendre–Gauss–Lobatto (LGL) points is used. These points also is rooted in Gaussian quadrature (with two degrees less accuracy) and include the endpoints of the interval [12,13]. These points are the roots of the Lobatto polynomials $L_{o_{n+1}}(\xi) = (1-\xi)P'_n(\xi)$. Although they are less accurate than Legendre–Gauss points, they must replace these points in some problems (such as mapping onto irregular intervals), and this slight loss of accuracy can be compensated by increasing the number of points.

3. Isoparametric formulation

After these mathematical preliminaries on Lagrange interpolation and its error analysis, we introduce the isoparametric formulation method based on this interpolation. First, we perform this in the one-dimensional case for the beam element, and afterwards we follow the details more extensively in the two-dimensional case for the quadrilateral and triangular elements.

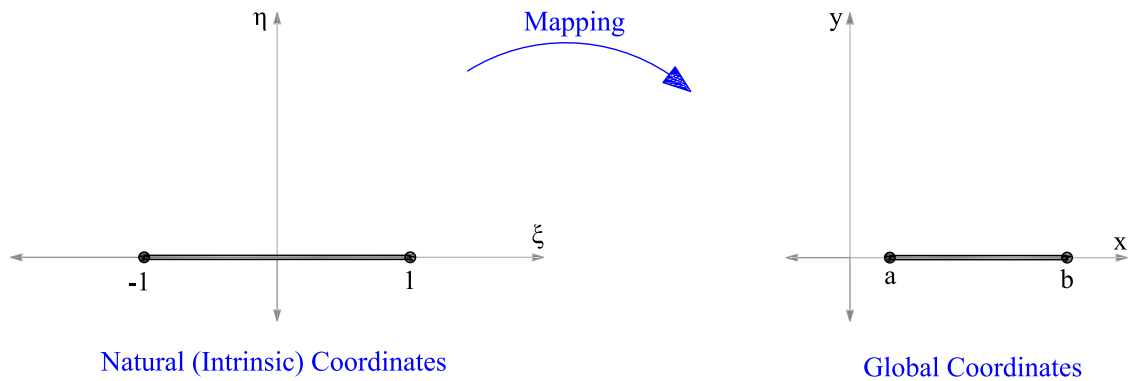


Fig. 1. Mapping of the line segment $[-1, 1]$ from natural coordinates to the line segment $[a, b]$ in global coordinates, with two fixed points chosen at the ends of the intervals.

3.1. Beam element (One-dimensional case)

To map natural coordinates to global coordinates in the one-dimensional case, we need to consider $x \in [a, b]$ as a function of $\xi \in [-1, 1]$. However, the nature of the function $x(\xi)$ depends on the type of mapping considered between these two coordinates. The Lagrange interpolation introduced earlier is a straightforward mapping that can provide a unique form of $x(\xi)$ by considering corresponding points between the natural and global coordinates (i.e., $x_i \leftrightarrow \xi_i$), mapping the entire natural interval $[-1, 1]$ to the global interval $[a, b]$ (Fig. 1). Therefore, if we consider $n+1$ fixed corresponding points, the function $x(\xi)$ will be uniquely constructed as an n -degree Lagrange interpolation as follows:

$$x(\xi) = \sum_{i=0}^n a_i \xi^i = \sum_{i=0}^n x(\xi_i) \mathcal{L}_{n,i}(\xi). \quad (6)$$

In the one-dimensional case, such a mapping is not strictly necessary, since by simply choosing the two endpoints the natural coordinates can be projected into the global coordinates. Because x represents a line segment defined by a linear equation, the higher-order coefficients in $x(\xi)$ vanish, and the interpolation reduces exactly to a first-degree polynomial that reproduces the line segment. This reflects the polynomial reproduction property of Lagrange interpolation: in the one-dimensional case, the interpolation exactly reproduces the linear polynomial representing the segment.

Moreover, the two chosen points (together with other potential points) need not necessarily be the endpoints of the interval. By considering the proportionality of the positions, corresponding points can be selected anywhere within the two intervals. Based on this, Legendre–Gauss points, which do not include the endpoints, can also be used, provided that the positions of the fixed points in the global coordinates are chosen proportionally to the Legendre–Gauss points in the natural coordinates. This can be achieved through a simple linear mapping (see Eq. (5)).

However, if the line in the global coordinates is curved, then more than two fixed points must be used to map the beam element onto it, and Eq. (6) must also be considered for the function $y(\xi)$:

$$y(\xi) = \sum_{i=0}^n a'_i \xi^i = \sum_{i=0}^n y(\xi_i) \mathcal{L}_{n,i}(\xi). \quad (7)$$

Thus, we encounter a one-dimensional to two-dimensional mapping.

Before formulating the parametric case for the two-dimensional situation, it should be recalled—as mentioned in the introduction—that the coefficient of each $x(\xi_i)$ and $y(\xi_i)$ in Eqs. (6) and (7) is referred to as the shape function of the i th point. These correspond to the displacement shape functions used in the finite element method to compute the displacement function u :

$$N_i^B(\xi) = \mathcal{L}_{n,i}(\xi) \quad \text{Beam element shape function} \quad (8)$$

Shape functions generally have two properties as follows [3]:

$$(1) : \sum_{i=0}^n N_i(\xi) = 1 \quad (9a)$$

$$(2) : N_i(\xi_j) = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases} \quad (9b)$$

It is clear that $N_i^B(\xi)$ satisfies both of the above criteria. Additionally, any other shape function introduced later will also have these two properties.

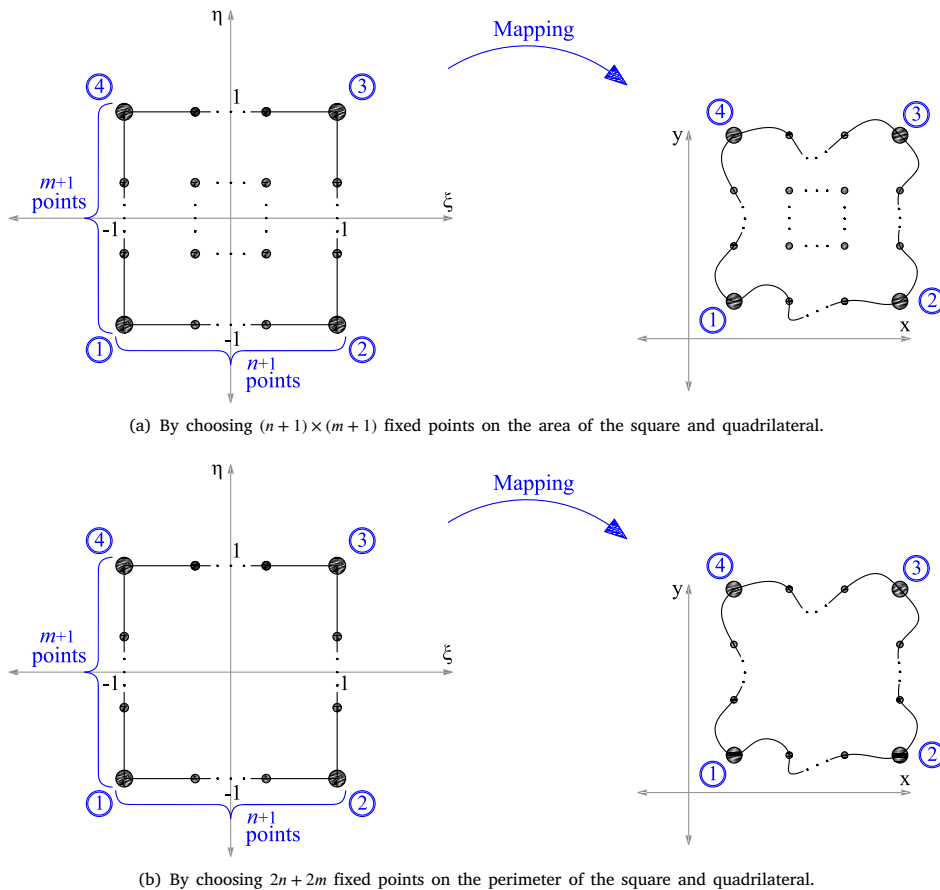


Fig. 2. Mapping of the reference square $\Omega_n = [-1, 1] \times [-1, 1]$ from natural coordinates to an arbitrary quadrilateral in global coordinates. Corresponding points are marked on the vertices so that the rest of the points are determined according to them.

3.2. Quadrilateral element (Two-dimensional case)

To map natural coordinates to global coordinates in the two-dimensional case, each of $x \in [a, b]$ and $y \in [c, d]$ must be considered as a function of $\xi \in [-1, 1]$ and $\eta \in [-1, 1]$. In this setting, the plane $\Omega_n := [-1, 1] \times [-1, 1]$ forms a square in the Cartesian coordinate system, which we call the reference square. As illustrated in Fig. 2, this reference square can be mapped onto any arbitrary quadrilateral in the global coordinates. If $n+1$ fixed points are chosen along ξ and $m+1$ fixed points along η , then the reference square will contain a total of $(n+1) \times (m+1)$ fixed points corresponding to the quadrilateral in the global domain.

As shown in Fig. 2(a), since for each fixed η_j , there are $n+1$ fixed ξ_i points along ξ , the function $x(\xi, \eta_j)$ (and similarly $y(\xi, \eta_j)$) will be unique if and only if it is a polynomial of degree n . We saw that this unique polynomial is the n -degree Lagrange interpolation. Therefore, we have:

$$x(\xi, \eta_j) = \sum_{i=0}^n a_i \xi^i = \sum_{i=0}^n x(\xi_i, \eta_j) \mathcal{L}_{n,i}(\xi). \quad (10)$$

Now, by generalizing this relationship for variable η using another Lagrange polynomial $\mathcal{M}_{m,j}(\eta)$, the unique polynomial $x(\xi, \eta)$ is obtained, which – along with $y(\xi, \eta)$ – maps the reference square $\Omega_n = [-1, 1] \times [-1, 1]$ in natural coordinates to an arbitrary quadrilateral in global coordinates:

$$x(\xi, \eta) = \sum_{i=0}^n \sum_{j=0}^m a_i \xi^i b_j \eta^j = \sum_{i=0}^n \sum_{j=0}^m x(\xi_i, \eta_j) \mathcal{L}_{n,i}(\xi) \mathcal{M}_{m,j}(\eta). \quad (11)$$

In this relationship, $x(\xi, \eta)$ is constructed using fixed points located both on the perimeter and inside the square. However, in isoparametric formulation, points known as the serendipity family, which are mainly located on the perimeter of the region, are usually used [4]. This way, the desired mapping, in addition to having a lower computational cost, does not require the correspondence of internal fixed points.

Therefore, with the reduction of fixed points, it is necessary to make changes to the polynomial $x(\xi, \eta)$ and remove some of its terms. Since adding a fixed point increases the polynomial degree by one, removing each fixed point should eliminate the higher-order terms in descending order. Thus, as shown in Fig. 2(b), by removing the internal fixed points of the square, the general form of the polynomial $x(\xi, \eta)$ in terms of $2n + 2m$ fixed points will be as follows:

$$x(\xi, \eta) = \sum_{i=0}^n \sum_{j=0}^m a_i \xi^i b_j \eta^j - \sum_{i=2}^n \sum_{j=2}^m a_i \xi^i b_j \eta^j. \quad (12)$$

As can be seen, the second part of this relationship is an incomplete polynomial; on the other hand, to write the equivalent Lagrange interpolation of this polynomial, it is necessary to write this part as a combination of complete polynomials. We have:

$$\begin{aligned} \sum_{i=2}^n \sum_{j=2}^m a_i \xi^i b_j \eta^j &= \left(\sum_{i=0}^n a_i \xi^i - (a_0 + a_1 \xi) \right) \left(\sum_{j=0}^m b_j \eta^j - (b_0 + b_1 \eta) \right) \\ &= \sum_{i=0}^n \sum_{j=0}^m a_i \xi^i b_j \eta^j - (a_0 + a_1 \xi) \sum_{j=0}^m b_j \eta^j - (b_0 + b_1 \eta) \sum_{i=0}^n a_i \xi^i + (a_0 + a_1 \xi)(b_0 + b_1 \eta). \end{aligned} \quad (13)$$

By substituting into the previous equation, we obtain:

$$x(\xi, \eta) = (a_0 + a_1 \xi) \sum_{j=0}^m b_j \eta^j + (b_0 + b_1 \eta) \sum_{i=0}^n a_i \xi^i - (a_0 + a_1 \xi)(b_0 + b_1 \eta). \quad (14)$$

Thus, the unique polynomial $x(\xi, \eta)$ on the boundary of the square $\xi\eta$ is obtained as follows:

$$\begin{aligned} x(\xi, \eta) &= \sum_{i=0, n}^m \sum_{j=0}^m x(\xi_i, \eta_j) \mathcal{L}_{1,i}(\xi) \mathcal{M}_{m,j}(\eta) + \sum_{i=0}^n \sum_{j=0, m} x(\xi_i, \eta_j) \mathcal{L}_{n,i}(\xi) \mathcal{M}_{1,j}(\eta) \\ &\quad - \sum_{i=0, n}^m \sum_{j=0, m} x(\xi_i, \eta_j) \mathcal{L}_{1,i}(\xi) \mathcal{M}_{1,j}(\eta). \end{aligned} \quad (15)$$

Accordingly, the general form of the quadrilateral shape function for each point on the boundary of the square is as follows:

$$\begin{aligned} N_{i,j}^Q(\xi, \eta) &:= \sum_{i=0}^1 \sum_{j=0}^m \mathcal{L}_{1,i}(\xi) \mathcal{M}_{m,j}(\eta) + \sum_{i=0}^n \sum_{j=0}^1 \mathcal{L}_{n,i}(\xi) \mathcal{M}_{1,j}(\eta) \\ &\quad - \sum_{i=0}^1 \sum_{j=0}^1 \mathcal{L}_{1,i}(\xi) \mathcal{M}_{1,j}(\eta). \end{aligned} \quad \text{Quadrilateral element shape function} \quad (16)$$

In this equation, note that $\mathcal{L}_{1,i}(\xi)$ and $\mathcal{M}_{1,j}(\eta)$ are first-degree Lagrange polynomials constructed from the fixed points -1 and $+1$, expressed as $(1-\xi)/2$, $(1+\xi)/2$, $(1-\eta)/2$, and $(1+\eta)/2$. By the polynomial reproduction property of Lagrange interpolation, on sides 1–2 and 3–4 (excluding the vertices), any $\mathcal{M}_{m,j}(\eta)$ restricted to $\eta \in \{-1, 1\}$ reduces to $\mathcal{M}_{1,j}(\eta)$; similarly, on sides 2–3 and 4–1, any $\mathcal{L}_{n,i}(\xi)$ restricted to $\xi \in \{-1, 1\}$ reduces to $\mathcal{L}_{1,i}(\xi)$. Consequently, for interior points along the sides, two of the three terms cancel and only one remains, whereas at the vertices all three terms are present.

As illustrated in Fig. 3, consider a quadrilateral with four equally spaced fixed points on each side. Based on the general form in Eq. (16), the cubic isoparametric formulation yields the following shape functions, matching [1,3,14]:

$$N_{i,j}^Q(\xi, \eta) = \frac{1}{32} (1 + \xi \xi_i) (1 + \eta \eta_j) [9(\xi^2 + \eta^2) - 10], \quad i = 0, 3, \quad j = 0, 3 \quad (17a)$$

$$N_{i,j}^Q(\xi, \eta) = \frac{9}{32} (1 + \xi \xi_i) (1 + 9\eta \eta_j) (1 - \eta^2), \quad i = 0, 3, \quad j = 1, 2 \quad (17b)$$

$$N_{i,j}^Q(\xi, \eta) = \frac{9}{32} (1 + \eta \eta_j) (1 + 9\xi \xi_i) (1 - \xi^2), \quad i = 1, 2, \quad j = 0, 3 \quad (17c)$$

In Fig. 3, side 1–2 is represented by a linear equation, sides 2–3 and 4–1 are represented by quadratic equations, and side 3–4 is represented by a cubic equation. Therefore, each side requires an appropriate number of fixed points to describe its geometry accurately, and the number of points on each side can be increased or decreased depending on the required accuracy.

Suppose side 1–2 contains $n' + 1$ points ($n' \leq n$) and side 2–3 contains $m' + 1$ points ($m' \leq m$). Then, the same as before by removing the other points on the boundary of the reference square, the polynomial form becomes:

$$\begin{aligned} x(\xi, \eta) &= b_0 \left(\sum_{i=0}^n a_i \xi^i - \sum_{i=0}^{n'} a_i \xi^i \right) + a_0 \left(\sum_{j=0}^m b_j \eta^j - \sum_{j=0}^{m'} b_j \eta^j \right) \\ &\quad + (b_0 + b_1 \eta) \sum_{i=0}^{n'} a_i \xi^i + (a_0 + a_1 \xi) \sum_{j=0}^{m'} b_j \eta^j - (a_0 + a_1 \xi)(b_0 + b_1 \eta). \end{aligned} \quad (18)$$

Accordingly, the equivalent Lagrange interpolation is expressed as:

$$x(\xi, \eta) = \left(\sum_{i=0}^n x(\xi_i, \eta_j) \mathcal{L}_{n,i}(\xi) - \sum_{i=0}^{n'} x(\xi_i, \eta_j) \mathcal{L}_{n',i}(\xi) \right) + \left(\sum_{j=0}^m x(\xi_i, \eta_j) \mathcal{M}_{m,j}(\eta) - \sum_{j=0}^{m'} x(\xi_i, \eta_j) \mathcal{M}_{m',j}(\eta) \right) \quad (19)$$

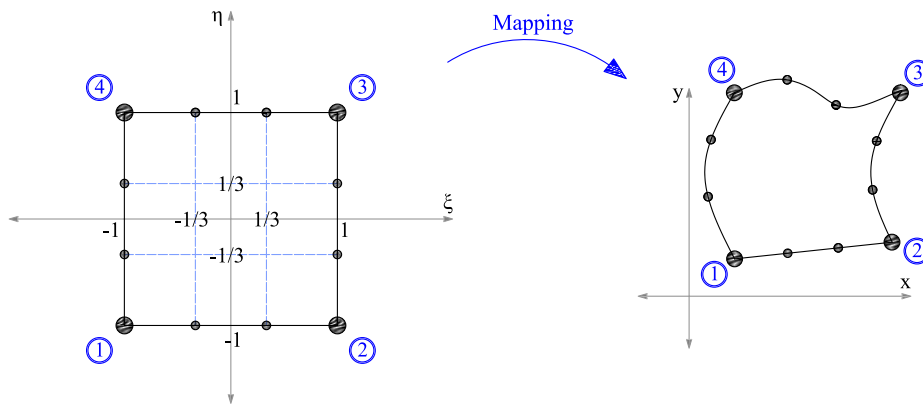


Fig. 3. Cubic isoparametric formulation for a quadrilateral with 4 fixed points on each side.

$$\begin{aligned}
 & + \sum_{i=0}^{n'} \sum_{j=0, m'} x(\xi_i, \eta_j) \mathcal{L}_{n', i}(\xi) \mathcal{M}_{1, j}(\eta) + \sum_{i=0, n'} \sum_{j=0}^{m'} x(\xi_i, \eta_j) \mathcal{L}_{1, i}(\xi) \mathcal{M}_{m', j}(\eta) \\
 & - \sum_{i=0, n} \sum_{j=0, m} x(\xi_i, \eta_j) \mathcal{L}_{1, i}(\xi) \mathcal{M}_{1, j}(\eta).
 \end{aligned}$$

Two situations can be distinguished:

- If $n' = n$ and $m' = m$, the formulation reduces to the symmetric standard form in Eq. (15).
- Otherwise, the location of the point on the quadrilateral must be considered, and according to the polynomial reproduction property of Lagrange interpolation, Eq. (19) changes as follows: * If the point lies on sides 1–2 or 2–3, which contain n' and m' fixed points respectively, then the bases $\mathcal{L}_{n', i}(\xi)$ and $\mathcal{M}_{m', j}(\eta)$ are replaced by $\mathcal{L}_{n', i}(\xi)$ and $\mathcal{M}_{m', j}(\eta)$. In this case, Eq. (15) is recovered, but with n, m replaced by n', m' . * If the point lies on sides 3–4 or 4–1, which contain n and m fixed points respectively, then one must examine how $\mathcal{L}_{n', i}(\xi)$ and $\mathcal{M}_{m', j}(\eta)$ —defined with fewer points—are constructed. From direct computation of the unknown coefficients in Eq. (18) and by induction, it is found that in this case the numerator of each $\mathcal{L}_{n', i}(\xi)$ and $\mathcal{M}_{m', j}(\eta)$ (see Eq. (2)) is defined in terms of n' and m' fixed points on the opposite side, while their denominators remain equal to those of $\mathcal{L}_{n, i}(\xi)$ and $\mathcal{M}_{m, j}(\eta)$.

Since this direct procedure is complicated and time-consuming, a more practical approach is to first introduce auxiliary points to raise n' and m' to n and m . Once Eq. (15) is re-established for the shape functions, the polynomial reproduction property of Lagrange interpolation can again be applied: for each point, depending on the maximum number of fixed points on its corresponding side, either n or n' and m or m' are substituted into the relation. Although the presence of auxiliary points presents the shape functions in a different form, this approach is simpler and more efficient than the fully direct method.

3.2.1. Shape functions related to Legendre–Gauss points

Since the Legendre–Gauss points do not include the vertices of the square, the connection between adjacent sides is lost, and only opposite sides remain correlated (see Fig. 4). Accordingly, the equation corresponding to the sum of points on sides 1–2 and 3–4 alone can describe $x(\xi, \eta)$ (denoted by the subscript $u-d$, for up–down):

$$x_{u-d}(\xi, \eta) = (b_0 + b_1 \eta) \sum_{i=0}^n a_i \xi^i = \sum_{i=0}^n \sum_{j=0}^1 x(\xi_i, \eta_j) \mathcal{L}_{n, i}(\xi) \mathcal{M}_{1, j}(\eta). \quad (20)$$

Similarly, the equation corresponding to the sum of points on sides 2–3 and 4–1 can also be used (denoted by the subscript $l-r$, for left–right):

$$x_{l-r}(\xi, \eta) = (a_0 + a_1 \xi) \sum_{j=0}^m b_j \eta^j = \sum_{i=0}^1 \sum_{j=0}^m x(\xi_i, \eta_j) \mathcal{L}_{1, i}(\xi) \mathcal{M}_{m, j}(\eta). \quad (21)$$

In each of these relations, the sum of the shape functions (the coefficients $x(\xi_i, \eta_j)$) independently equals one, which demonstrates the independence of the two formulations. The main difficulty is that each relation represents only the actual geometry of two opposite sides, while the other two sides are treated as straight lines. If those sides are indeed straight, no issue arises; otherwise, these relations are not valid for the entire quadrilateral. Averaging the two relations is also not a correct solution, since it would mix the true geometry of each side with a straight line, thereby distorting the actual shape.

To resolve this problem, we assume that the Gauss points on the sides of the square are pulled inward, i.e., placed on the largest inner square shown in Fig. 4. Under this assumption, the fundamental polynomial of the serendipity elements (Eq. (14)), which is defined for the whole square, remains valid for these points. Moreover, since shape functions are not unique and only need to satisfy the essential properties listed in Eq. (9) (partition of unity and Kronecker delta property) [3], one can, inspired by Eq. (14), rewrite x_{u-d} and x_{l-r} so that their sum equals one over the entire square.

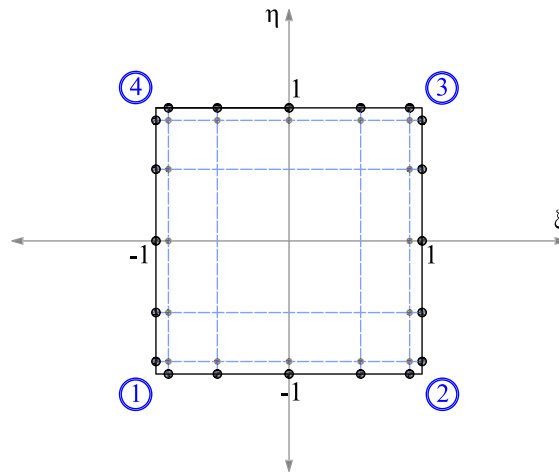


Fig. 4. Distribution of five Gauss points on the reference square $\Omega_n = [-1, 1] \times [-1, 1]$. The smaller circles on the largest inner square are introduced to establish connectivity between adjacent sides.

Careful inspection of Eq. (14) shows that its first and second terms separately correspond to opposite sides, while the third term can simultaneously apply to both pairs of opposite sides. Therefore, this third term is symmetrically divided into two halves, each added to one of the independent relations. Consequently, we obtain:

$$x_{u-d}(\xi, \eta) = (b_0 + b_1 \eta) \sum_{i=0}^n a_i \xi^i - \frac{1}{2} (a_0 + a_1 \xi) (b_0 + b_1 \eta) \quad (22)$$

$$= \sum_{i=0}^n \sum_{j=0}^1 x(\xi_i, \eta_j) \mathcal{L}_{n,i}(\xi) \mathcal{M}_{1,j}(\eta) - \frac{1}{2} \sum_{i=0, n}^1 \sum_{j=0}^1 x(\xi_i, \eta_j) \mathcal{L}_{1,i}(\xi) \mathcal{M}_{1,j}(\eta),$$

$$x_{l-r}(\xi, \eta) = (a_0 + a_1 \xi) \sum_{j=0}^m b_j \eta^j - \frac{1}{2} (a_0 + a_1 \xi) (b_0 + b_1 \eta) \quad (23)$$

$$= \sum_{i=0}^1 \sum_{j=0}^m x(\xi_i, \eta_j) \mathcal{L}_{1,i}(\xi) \mathcal{M}_{m,j}(\eta) - \frac{1}{2} \sum_{i=0, 1}^1 \sum_{j=0, m}^m x(\xi_i, \eta_j) \mathcal{L}_{1,i}(\xi) \mathcal{M}_{1,j}(\eta).$$

It should be noted that in $x_{u-d}(\xi, \eta)$ the index η_j can only take the boundary values -1 and $+1$, while ξ_i corresponds to Gauss points along the ξ axis. Conversely, in $x_{l-r}(\xi, \eta)$ the index ξ_i can only take the boundary values -1 and $+1$, while η_j corresponds to Gauss points along the η axis. Thus, in each relation only one of the two variables is sampled at Gauss points, whereas the other is fixed at the boundary values $\{-1, +1\}$.

Thus, although the connection between adjacent sides is still not established, the shape functions are now designed so that their sum equals one over the entire square. Therefore, Legendre–Gauss points, which provide high accuracy in interpolation, can consistently be used in isoparametric formulations.

At the end of this section, it should be emphasized that the general rule of isoparametric formulation can also be extended to higher dimensions in the same manner as described for the two-dimensional case. In other words, similar to Eq. (11), one first writes the initial polynomial; then, by eliminating the interior points of the boundary under consideration, a polynomial based on serendipity points is obtained; and finally, by establishing the correspondence between equivalent points in the two coordinate systems, the Lagrange interpolation equivalent to this polynomial is derived.

3.3. Triangular element (Two-dimensional case)

Now, if we want to consider a triangular element in global coordinates as an image of a regular shape in natural coordinates, we will encounter more complex shape functions. If, as shown in Fig. 5, we consider the reference triangle as a right-angled isosceles triangle with $\xi \in [0, 1]$ and $\eta \in [0, 1]$ and $n + 1$ fixed points on each leg, then similar to the square case, the function $x(\xi, \eta)$ will be a set of equations on each fixed η_j as follows:

$$x(\xi, \eta) = b_0 \sum_{i=0}^n a_i \xi^i + b_1 \eta \sum_{i=0}^{n-1} a_i \xi^i + \dots + b_{n-1} \eta^{n-1} (a_0 + a_1 \xi) + b_n \eta^n a_0. \quad (24)$$

If we also consider the set of equations on each fixed ξ_i , we will reach the same unique expression. Note that the number of fixed points on ξ and η must always be the same, as both are considered projections of points on the hypotenuse. However, it is noteworthy that if the interior points are removed, the above expression will no longer be unique for defining $x(\xi, \eta)$; hence, the serendipity points for the triangle also include interior points.

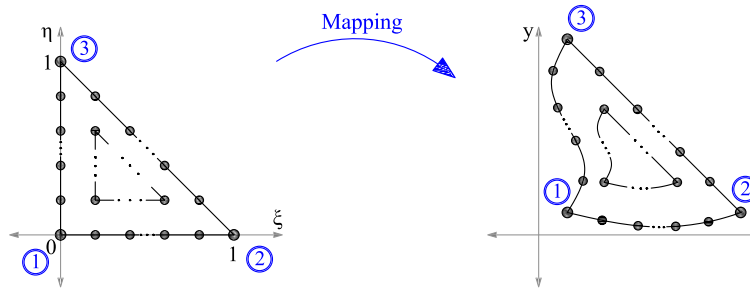


Fig. 5. Mapping of the reference triangular element from natural coordinates to an arbitrary triangular element in global coordinates, with the selection of $(n+1)(n+2)/2$ fixed points on the boundary of the triangle and the triangular element.

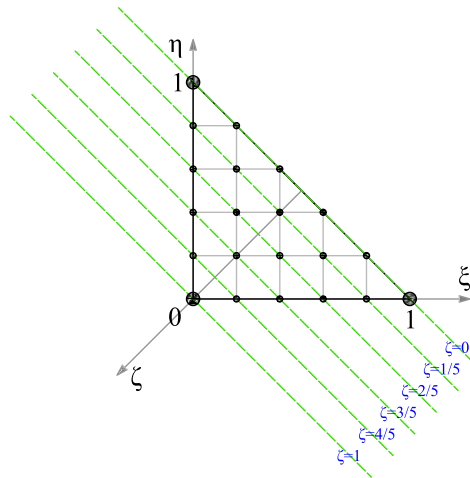


Fig. 6. Representation of the lines ζ_i on the ζ axis passing through equally spaced fixed points of the reference triangle.

Since Eq. (24) is complex and cannot easily be extracted as a Lagrange interpolation, we define a new variable $\zeta := 1 - \xi - \eta$ to do this.

As seen in Fig. 6, the entire triangle can be interpolated using a one-dimensional interpolation with the Lagrange function $\mathcal{O}_{n,i}(\zeta)$. However, since for each fixed ζ_i (except $\zeta_0 = 1$ which passes through the vertex of the triangle) there is more than one point where $\mathcal{O}_{n,i}(\zeta)$ equals one, this one-dimensional interpolation alone cannot satisfy the shape function criteria. Instead, by combining this Lagrange function with Lagrange functions dependent on ξ and η , we consider the shape function for each point as follows, which will meet both shape function criteria:

$$N_{i,j}^T(\xi, \eta) := \mathcal{L}_{i+1,i}(\xi) \mathcal{M}_{j+1,j}(\eta) \mathcal{O}_{n-i-j,0}(\zeta). \quad \text{Triangular element shape function} \quad (25)$$

However, it is important to note that this shape function is only defined for Equidistant (Eq) points and therefore cannot be used for Gaussian points. Although attempts have been made to use Gaussian points for triangular mapping [14], these approximations are still not as robust and strong as the isoparametric formulation for quadrilateral elements. This highlights the advantage and superiority of quadrilateral mapping. Additionally, the isoparametric formulation for quadrilateral elements does not require the use of interior points, which simplifies point placement and reduces computations. Given the simplicity of meshing with triangles compared to rectangles and the lack of application of triangles in spectral methods, the necessity of triangular mapping is reduced. In the following, we will show with a practical example that includes a side with sharp points, that any shape in global coordinates can be considered as a quadrilateral by defining 4 sides on its boundary, and thus the mapping for this element can be used for any arbitrary two-dimensional shape.

4. Derivatives of the displacement function based on natural coordinates

Since the main purpose of the isoparametric formulation is to regularize the computational domain for solving differential equations, it is necessary to exploit this regular natural space by defining the displacement function $u(x, y)$ and its derivatives—all of which are initially unknown—in natural coordinates and treating them as direct functions of ξ and η (i.e., $u(\xi, \eta)$). However,

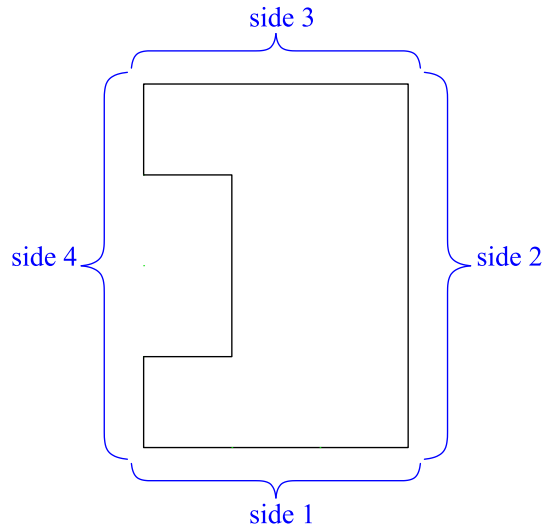


Fig. 7. Bracket-shaped quadrilateral.

because the differential equation itself is defined in global coordinates, the unknown derivative du/dx must be obtained using the chain rule in terms of the known derivative $du/d\xi$:

$$\frac{du}{d\xi} = \frac{du}{dx} \frac{dx}{d\xi}, \quad |J| := \frac{dx}{d\xi} \Rightarrow \frac{du}{dx} = \frac{1}{|J|} \left(\frac{du}{d\xi} \right). \quad (26)$$

In Eq. (26), note that J is the Jacobian, which shows the derivative of the primary coordinates with respect to the secondary coordinates. It can also be shown that higher-order derivatives will be as follows:

$$\frac{d^n u}{dx^n} = \frac{1}{|J|^n} \frac{d^n u}{d\xi^n}. \quad (27)$$

In the two-dimensional case, similarly, we obtain $\partial u/\partial x$ and $\partial u/\partial y$ using the chain rule in terms of $\partial u/\partial \xi$ and $\partial u/\partial \eta$, resulting in:

$$\frac{\partial u}{\partial x} = \frac{1}{|J|} \begin{vmatrix} \frac{\partial u}{\partial \xi} & \frac{\partial u}{\partial \eta} \end{vmatrix}, \quad \frac{\partial u}{\partial y} = \frac{1}{|J|} \begin{vmatrix} \frac{\partial u}{\partial \xi} & \frac{\partial u}{\partial \eta} \end{vmatrix}, \quad |J| = \begin{vmatrix} \frac{\partial x}{\partial \xi} & \frac{\partial y}{\partial \xi} \\ \frac{\partial x}{\partial \eta} & \frac{\partial y}{\partial \eta} \end{vmatrix}. \quad (28)$$

For higher-order derivatives, we also have:

$$\frac{\partial^n u}{\partial x^n} = \left\{ \frac{1}{|J|} \left[\frac{\partial y}{\partial \eta} \frac{\partial}{\partial \xi} - \frac{\partial y}{\partial \xi} \frac{\partial}{\partial \eta} \right] \right\}^n u, \quad (29a)$$

$$\frac{\partial^m u}{\partial y^m} = \left\{ \frac{1}{|J|} \left[\frac{\partial x}{\partial \xi} \frac{\partial}{\partial \eta} - \frac{\partial x}{\partial \eta} \frac{\partial}{\partial \xi} \right] \right\}^m u, \quad (29b)$$

$$\frac{\partial^m}{\partial y^m} \frac{\partial^n u}{\partial x^n} = \left\{ \frac{1}{|J|} \left[\frac{\partial x}{\partial \xi} \frac{\partial}{\partial \eta} - \frac{\partial x}{\partial \eta} \frac{\partial}{\partial \xi} \right] \right\}^m \left\{ \frac{1}{|J|} \left[\frac{\partial y}{\partial \eta} \frac{\partial}{\partial \xi} - \frac{\partial y}{\partial \xi} \frac{\partial}{\partial \eta} \right] \right\}^n u. \quad (29c)$$

5. Inverse mapping from global to natural coordinates

Before concluding, it is appropriate to mention that to examine boundary conditions on shared boundaries, it is necessary to determine ξ and η in terms of the common x and y for the other region. Since the functions $x(\xi, \eta)$ and $y(\xi, \eta)$ are nonlinear (except in the two-point case), to obtain ξ and η , we must solve a system of two nonlinear equations with two unknowns. If these common points are x_0 and y_0 , this system of equations is:

$$\begin{cases} x(\xi, \eta) = x_0 \\ y(\xi, \eta) = y_0 \end{cases} \quad (30)$$

This nonlinear system can be solved using iterative methods such as the Newton method. However, it should be noted that since $x(\xi, \eta)$ and $y(\xi, \eta)$ are not one-to-one (except in the two-point case), a good initial guess for ξ and η based on the existing geometry is necessary to achieve an accurate solution.

Table 1

Average relative error of the isoparametric formulation on the boundary of the quadrilateral for different numbers of fixed points on the indented side.

Number of points	10	20	30
Eq points	8.69×10^{-2}	6.54×10^{-1}	3.64×10^2
CG points	9.11×10^{-2}	4.95×10^{-3}	4.90×10^{-4}
LGL points	9.32×10^{-2}	1.90×10^{-3}	7.44×10^{-4}
LG points	7.71×10^{-2}	9.11×10^{-3}	2.82×10^{-3}

Table 2

Average relative error of the isoparametric formulation on the fourth side for different numbers of fixed points on this side.

Number of points	10	20	30
Eq points	3.47×10^{-1}	2.62	1.45×10^2
CG points	3.64×10^{-1}	1.98×10^{-2}	1.96×10^{-3}
LGL points	3.73×10^{-1}	7.59×10^{-3}	2.97×10^{-3}
LG points	2.56×10^{-1}	1.69×10^{-2}	3.50×10^{-3}

6. Example and discussion

In this section, we aim to use the isoparametric formulation method to map the reference square $\Omega_n = [-1, 1] \times [-1, 1]$ in natural coordinates onto a quadrilateral, part of which is described by a piecewise function. Fig. 7, which could represent the chamber of a nuclear reactor [3] or a simulation of a unit cell surrounding a macromolecule [15], has the advantage that its left side is described not by a polynomial function of degree n , but by a piecewise function. If this side can be described by continuous functions $x(\xi, \eta)$ and $y(\xi, \eta)$, the power of the isoparametric formulation becomes more evident.

In our previous work [15], when solving the Poisson–Boltzmann equation in a bracket-shaped region using the pseudo-spectral method, we divided the domain into five rectangles. This subdivision allowed each rectangular sub-region to be mapped onto the reference square, keeping a single differential equation valid in each sub-region and ensuring that each side of the rectangle was governed by one boundary condition rather than multiple different conditions on separate segments.

However, here we consider this bracket-shaped region directly as an projection of the reference square, so that using the isoparametric formulation, the functions $x(\xi, \eta)$ and $y(\xi, \eta)$ can reconstruct this shape. Consequently, for solving the equation, we only need to mesh one region instead of five sub-regions, which simplifies the process significantly. We have also slightly changed the previous scale to increase the indentation and misalignment of the boundaries, allowing us to better assess the efficiency of the isoparametric formulation.

Fig. 8 shows the results of this formulation for 10 fixed points on the fourth side. Given the linear shape of sides 1 to 3, we have only considered their endpoints as fixed points to increase the program's execution speed. However, for the fourth side, we need to consider more points to approximate this non-differentiable piecewise function continuously. In this regard, since this side itself consists of five line segments, we have considered fixed points on each segment proportional to their length relative to the total length of the side. It is worth mentioning that these fixed points are also distributed on their respective line segments according to the type of point distribution in natural coordinates.

This figure also shows the distribution of interior points of the quadrilateral for several fixed ξ . As observed, the isoparametric formulation based on Legendre–Gauss points ensures that the projection of the interior points of the reference square Ω_n lies entirely within the quadrilateral. In contrast, this formulation based on two other Gaussian distributions does not achieve such success. The inefficiency of equispaced fixed points is also evident; the resulting shape clearly demonstrates the Runge phenomenon [16], i.e., oscillation between fixed points.

However, this is not the whole story. As shown in Tables 1 and 2, increasing the number of fixed points on the fourth side results in the Chebyshev–Gauss and Legendre–Gauss–Lobatto distributions yielding the best results for the isoparametric formulation. In contrast, for fewer fixed points, the Legendre–Gauss distribution provides the best results for this formulation. But in general, the increase in accuracy for increasing the number of fixed points based on Gaussian distributions causes the isoparametric formulation to reproduce Fig. 7 exactly. It is also worth noting that the reason for separately examining the average relative error on the fourth side is the increased likelihood of mapping errors in uneven regions.

7. Conclusion

The isoparametric formulation, which is designed to smooth the boundaries of the solution space of a differential equation, can be highly accurate when it uses Gaussian points as the corresponding fixed points between the natural and global coordinates. Accordingly, for two-dimensional shapes, the quadrilateral mapping is preferable to the triangular mapping; because it can both use Gaussian points and does not require the definition of internal fixed points. The final example, which has a side with sharp points, reveals the importance and value of this formulation, especially for spectral methods. It should be noted that although

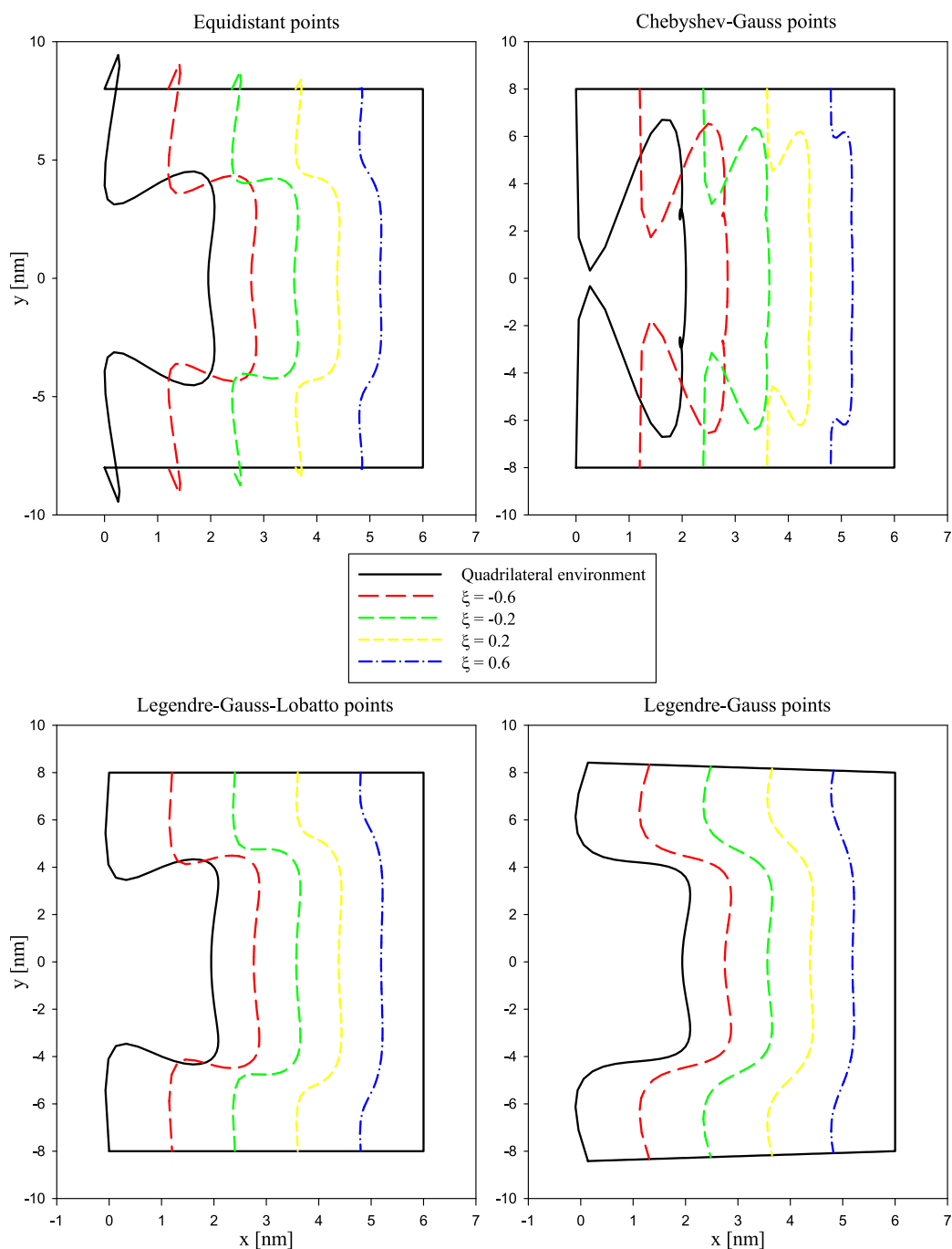


Fig. 8. Mapping of the reference square from natural coordinates to global coordinates to obtain the bracket-shaped quadrilateral, using 10 fixed points on the indented side distributed in four ways.

increasing the number of fixed points leads to better results for the isoparametric formulation, it also increases the program's execution time. On the other hand, for some simulated shapes where only the overall shape is considered, achieving high accuracy is not necessary. Therefore, using the Legendre–Gauss distribution, which provides higher accuracy with fewer fixed points compared to other distributions, is desirable for the isoparametric formulation.

To the best of our knowledge, our approach presents a novel perspective in leveraging Legendre–Gauss points for interpolation-based mapping in quadrilateral elements, and it offers an efficient alternative for overcoming the limitations associated with triangular mappings.

CRediT authorship contribution statement

Esmail Motevali: Writing – review & editing, Writing – original draft, Software, Methodology, Data curation, Conceptualization.
Houshyar Noshad: Supervision.

Declaration of competing interest

The authors declared that they have no conflict of interest.

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